

● HARD

● INFORMATION AND IDEAS

Mixed

10 questions · ~15 min

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Q1

INFORMATION AND IDEAS

The Uto-Aztecan language family is divided into a northern branch, which includes the Shoshone language of present-day Idaho and Utah, and a southern one, whose best-known representative is Nahuatl, the language of the Aztec Empire in Mexico. Lexical similarities across the family, including of botanical terms, confirm descent from a single language spoken millennia ago, and the family's geographical distribution suggests an origin in what is now the US Southwest. However, vocabulary pertaining to maize isn't shared between northern and southern branches, despite the crop's universal cultivation among Uto-Aztecan tribes. Given archaeological evidence that maize originated in Mexico and diffused northward into what became the US Southwest, some linguists reason that _____blank

Which choice most logically completes the text?

- A. northern Uto-Aztecan tribes likely obtained the crop directly from a southern Uto-Aztecan tribe rather than from a non-Uto-Aztecan tribe.
- B. variation in maize-related vocabulary within each branch of the Uto-Aztecan family likely reflects regionally specific methods for cultivating the crop.
- C. southern Uto-Aztecan tribes likely acquired maize at roughly the same time as northern Uto-Aztecan tribes did, though from different sources.
- D. the family's division into northern and southern branches likely preceded the acquisition of the crop by the Uto-Aztecan tribes.

For centuries, the widespread acknowledgment of the involvement of the cerebellum—a dense brain structure in vertebrates—in coordinating motor control in humans has hindered recognition of other possible functions of the structure. Neuroscience research from the last two decades now suggests that the cerebellum regulates emotion and social behavior, and recent research by Ilaria Carta and colleagues has identified a pathway connecting the cerebellum to a center for motivation and reward processing known as the ventral tegmental area (VTA).

Which choice best states the main idea of the text?

- A. The recent verification of a pathway between the VTA and the cerebellum confirms the cerebellum's long-suspected role in motor coordination.
- B. Recent advances in the field of neuroscience have challenged widely accepted claims about the function of a pathway connecting the VTA and the cerebellum.
- C. The cerebellum has primarily been thought to regulate motor functioning, but in recent years neuroscience researchers have been uncovering additional functions.
- D. Technological limitations have historically hindered the study of the cerebellum, but the recent development of new technologies has led to greater insights into its functions.

In 2018, scientists discovered an immense aggregation of *Muusoctopus robustus* (pearl octopuses) along a hydrothermal vent 3,200 meters beneath the ocean's surface. Water temperatures at this site—named the Octopus Garden—climb as high as 11°C, much warmer than the ambient 1.6°C typical at this depth. Based on observations made over three years, scientists concluded that temperatures at the site likely confer reproductive benefits and that the site is used exclusively for reproduction—6,000 *M. robustus* adults, hatchlings, and eggs were observed at the garden, but no juveniles were present.

Which statement about *M. robustus* and the Octopus Garden is best supported by the text?

- A. *M. robustus* leave the Octopus Garden upon reaching an intermediary stage of development.
- B. The *M. robustus* population at the Octopus Garden remains stable despite variations in water temperature.
- C. *M. robustus* nests in the Octopus Garden contain on average fewer but larger eggs than nests at similar ocean depths.
- D. The Octopus Garden provides an ideal feeding ground for *M. robustus* hatchlings.

O Pioneers! is a 1913 novel by Willa Cather. In the novel, Cather depicts Alexandra Bergson as a person who takes comfort in understanding the world around her:
_____ blank

Which quotation from *O Pioneers!* most effectively illustrates the claim?

- A. “She looked fixedly up the bleak street as if she were gathering her strength to face something, as if she were trying with all her might to grasp a situation which, no matter how painful, must be met and dealt with somehow.”
- B. “She had never known before how much the country meant to her. The chirping of the insects down in the long grass had been like the sweetest music. She had felt as if her heart were hiding down there, somewhere, with the quail and the plover and all the little wild things that crooned or buzzed in the sun. Under the long shaggy ridges, she felt the future stirring.”
- C. “Alexandra drove off alone. The rattle of her wagon was lost in the howling of the wind, but her lantern, held firmly between her feet, made a moving point of light along the highway, going deeper and deeper into the dark country.”
- D. “Alexandra drew her shawl closer about her and stood leaning against the frame of the mill, looking at the stars which glittered so keenly through the frosty autumn air. She always loved to watch them, to think of their vastness and distance, and of their ordered march. It fortified her to reflect upon the great operations of nature, and when she thought of the law that lay behind them, she felt a sense of personal security.”

While researching a topic, a student has taken the following notes:

- In a 2023 study, environmental scientist Jazmin Locke-Rodriguez and colleagues tested the use of floating treatment wetlands (FTWs) in Florida.
- FTWs are artificial floating platforms of plants used to remediate polluted or nutrient-imbalanced water.
- Finding: FTWs using marigold flowers removed 52% more total phosphorus than the control.
- Finding: The test yielded 65 market-quality blooms per square meter.
- The authors concluded marigolds showed “promising potential as a commercially viable remediating crop cultivated on FTWs in South Florida.”

The student wants to present the findings of the study. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. The authors concluded that marigolds grown on FTWs were “commercially viable,” having produced 65 blooms per square meter of market-quality blooms in a 2023 study.
- B. In a 2023 study, Locke-Rodriguez and colleagues found that marigolds cultivated on FTWs produced 52% more market-quality flower blooms than the control.
- C. Locke-Rodriguez and colleagues found that FTWs using marigolds not only helped remove phosphorus from the water but also yielded market-quality blooms.
- D. FTWs using marigolds, Locke-Rodriguez and colleagues found, yielded 65 flower blooms and removed 52% of phosphorus from the water.

If the formation of Earth’s mantle had been purely a product of core differentiation—whereby heavier elements sink toward the core and lighter elements rise—the upper mantle would be depleted of heavy siderophile elements. Siderophiles are much more abundant in the mantle than predicted in that model, however.

_____blank extraterrestrial material containing siderophiles, likely from asteroid or comet impacts, almost certainly accreted to Earth following core differentiation.

Which choice completes the text with the most logical transition?

- A. That said,
- B. Hence,
- C. For example,
- D. Likewise,

In modern plays, actors typically won't acknowledge the _____ blank do so breaks the fourth wall, a metaphorical barrier between actors and audiences that allows viewers to suspend the knowledge that they're watching a staged performance.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. audience. As to
- B. audience to
- C. audience. To
- D. audience, to

Quantum particles of light—photons—provide an unhackable means of transmitting encryption keys over networks, as attempts to observe particles in quantum states will invariably alter the particles _____blank dismantle any information they transmit.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. and in the process,
- B. and, in the process,
- C. and in the process—
- D. and, in the process

Containing over 160 billion base pairs of DNA resulting from polyploidy (whole genome duplication) and accumulation of noncoding DNA without significant deletion, the genome of *Tmesipteris oblongeolata* (New Caledonian fork fern) is the largest known among eukaryotes—and, given the high biological costs of extremely large genomes, is unlikely to be _____ blank.

Which choice completes the text with the most logical and precise word or phrase?

- A. protracted
- B. superseded
- C. curtailed
- D. obviated

Political blogs with conspicuous ideological alignments became an integral component of US media in the early 2000s. While some commentators lauded this development, asserting that such blogs had a welcome transparency missing from traditional news, less _____blank observers countered that such blogs tended to ideological extremes that exacerbated political polarization to problematic levels.

Which choice completes the text with the most logical and precise word or phrase?

- A. sanguine
- B. recalcitrant
- C. misanthropic
- D. earnest